

AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

October 21, 2015 1:00-2:50 PM

Close Books/Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (30) Suppose that a spacecraft with mass $m=100\text{kg}$ operates near a circular orbit with orbital radius $r=6,578\text{km}$. The Clohessy-Wiltshire equations can describe its orbital perturbation dynamics in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. Under some conditions, the Clohessy-Wiltshire equations can be further approximated by

$$m\Delta\ddot{x} = F_x$$

$$m\Delta\ddot{y} = F_y$$

$$m\Delta\ddot{z} = F_z$$

$$\begin{aligned} 100\Delta\ddot{x} &= F_x \\ 100\Delta\ddot{y} &= F_y \\ 100\Delta\ddot{z} &= F_z \end{aligned}$$

Here $(\Delta x, \Delta y, \Delta z)$ are perturbations of the spacecraft position from the circular orbit, and (F_x, F_y, F_z) are propulsive force components expressed in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. This approximation is often acceptable over a sufficiently small time period $T > 0$.

- a. (15) Estimate how small the maneuver time period T should be so that the above equations are a satisfactory approximation of the Clohessy-Wiltshire equations within accuracy of about 3%. Hint: Consider the following analytic solutions to the Clohessy-Wiltshire equations:

$$x(t) = \frac{\dot{x}_0}{\omega} \sin(\omega t) - \left(3x_0 + \frac{2\dot{y}_0}{\omega}\right) \cos(\omega t) + \left(4x_0 + \frac{2\dot{y}_0}{\omega}\right)$$

$$y(t) = \left(6x_0 + \frac{4\dot{y}_0}{\omega}\right) \sin(\omega t) + \frac{2\dot{x}_0}{\omega} \cos(\omega t) - (6\omega x_0 + 3\dot{y}_0)t + \left(y_0 - \frac{2\dot{x}_0}{\omega}\right)$$

$$z(t) = \frac{\dot{z}_0}{\omega} \sin(\omega t) + z_0 \cos(\omega t)$$

$$T = \frac{2\pi}{\omega}$$

$$0.03T = 0.03 \frac{2\pi}{\omega}$$

- b. (15) Using the above approximation, a two-impulse rendezvous maneuver is to transfer the initial spacecraft position vector $\Delta\vec{r}(0) = -10\hat{i}\text{km}$ and initial velocity vector $\Delta\vec{v}(0) = 0$ to the terminal spacecraft position vector $\Delta\vec{r}(60\text{sec}) = 0$ and terminal velocity vector $\Delta\vec{v}(60\text{sec}) = 0$. What is the change in spacecraft velocity vector that occurs at each of the impulsive burns?

$$\omega = \sqrt{\frac{\mu}{R^3}}$$

$$T = \frac{2\pi}{\omega}$$

$$\omega^2 x = F_x$$

$$\omega \rightarrow 1 \quad T = \frac{2\pi}{\omega}$$

$$\frac{6\pi}{100}$$

$$C_0 \quad y_0 = -\frac{2\dot{x}_0}{\omega}$$

$$v = 0.03$$

$$\Delta\vec{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

2. (40) Consider the equations of motion for the Earth-Moon circular restricted three-body problem:

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_E (x - D_E)}{r_E^3} - \frac{\mu_M (x + D_M)}{r_M^3} + \frac{F_x}{m} \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m} \end{cases}, \begin{cases} r_E^2 = (x - D_E)^2 + y^2 + z^2 \\ r_M^2 = (x + D_M)^2 + y^2 + z^2 \\ \mu_E = 398,601 \text{ km}^3 / \text{sec}^2 \\ \mu_M = 4,887 \text{ km}^3 / \text{sec}^2 \\ D_E = 4,674 \text{ km} \\ D_M = 380,073 \text{ km} \\ \omega_0 = 2.66199 \times 10^{-6} \text{ rad/sec} \end{cases}$$

- (5) Show that $L_4 : (\bar{x} = -187,699.3 \text{ km}, \bar{y} = 333,201.5 \text{ km}, \bar{z} = 0)$ is a Lagrangian point.
- (20) Linearize the above equations of motion with respect to the Lagrangian point given in (a).
- (5) What are the eigenvalues of the linearized out-of-plane equations of motion?
- (10) Discuss the stability of out-of-plane motion based on its eigenvalues.

3. (20) For the 3-2-1 rotation, derive the angular velocity components $\vec{\omega} = [\omega_x, \omega_y, \omega_z]^T$ in terms of the Euler angles (ψ, θ, ϕ) and their rates.

4. (10) Discuss the following concepts briefly and precisely.

- (5) Halo Orbit (Original Definition in Celestial Mechanics)

- (5) Exponential Coordinates

enc

Δ

μ